Roots of polynomial functions

1

a No **b** Yes **c** Yes **d** No

2

a

i $x = -6,\ x = 3,\ x = -9$

ii $x = -6$ has the multiplicity of 3, $x = 3$ has the multiplicity of 1 and $x = -9$ has the multiplicity of 1.

b

i $x = -1,\ x = 6,\ x = -2$

ii $x = -1$ has the multiplicity of 2, $x = 6$ has the multiplicity of 1 and $x = -2$ has the multiplicity of 5.

c

i $x = 6,\ x = -2,\ x = 0$

ii $x = 6$ has the multiplicity of 4, $x = -2$ has the multiplicity of 5 and $x = 0$ has the multiplicity of 3.

d

i $x = 3,\ x = \sqrt{5},\ x = -\sqrt{5}$

ii $x = 3$ has the multiplicity of 1, $x = \sqrt{5}$ has the multiplicity of 1 and $x = -\sqrt{5}$ has the multiplicity of 1.

3

a Yes **b** No **c** No **d** No

4

a Yes **b** Yes **c** Yes **d** No

e Yes **f** No **g** No **h** Yes

5

a $x + 4,\ x + 3,\ x - 2,\ x - 4,\ x - 5$

b $h(x) = (x + 4)(x + 3)(x - 2)(x - 4)(x - 5)$

c

i Yes **ii** No **iii** No **iv** Yes

d 0

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Equations of polynomial functions

6

a $x = -1, -5$

b $x = -1$

c $y = (x+1)^2 (x+5)$

7 $y = (x+4)^2 (x-1)$

8

a $y = -4x^3 + 12x^2 + 52x - 60$

b $y = -4x^3 + 28x^2 - 40x$

c $y = \frac{1}{4}x^3 - 9x$

d $y = 3x^3 + 6x^2 - 57x - 60$

e $y = 3x^3 - 30x^2 + 87x - 60$

f $y = -2x^3 - 4x^2 + 22x + 24$

g $y = 5x^3 + 40x^2 + 95x + 60$

h $y = -5x^3 + 15x^2 + 30x - 40$

9 $b = -5$

10

a $y = 2(x+3)(x+1)(x-1)$

b $y = -(x+4)(x-2)(x+1)$

c $y = (x+1)(x-2)(x-4)$

d $y = -2(x+1)(x-1)(x-3)$

e $y = -\frac{1}{6}(x+4)(x+3)(x+2)(x+1)$

f $y = 2(x+2)(x+1)(x-1)(x-2)$

g $y = \frac{1}{4}(x-1)(x-2)(x-3)(x-4)$

h $y = -2(x+2)(x+1)(x-1)(x-2)$

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